INCH-POUND

MIL-S-52408C(ME)
19 December 1989
SUPERSEDING
MIL-S-52408B(ME)
1 July 1983

MILITARY SPECIFICATION

SPREADER, SLING: TELESCOPIC, VEHICLE LIFTING, 10-TON CAPACITY

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tubular-steel telescopic sling spreader.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

FF-P-386	- Pins, Cotter (Split).
FF-W-92	- Washer, Flat (Plain).
QQ-B-637	- Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3940
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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RR-C-271	- Chains and Attachments Welded and Weldless.
PPP-B-601	- Boxes, Wood Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packing and Marking.
MS20392	- Pin, Straight. Headed - Drilled Shank.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1	- Surface Texture (Surface Roughness, Waviness And Lay).

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ANSI/ASTM A 36	- Structural Steel.
ANSI/ASTM A 322	- Steel Bars, Alloy, Standard Grades.
ANSI/ASTM A 513	- Electric-Resistance-Welded Carbon and Alloy Steel Mechanical Tubing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103)

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AMERICAN WELDING SOCIETY, INC. (AWS)

ANSI/AWS A2.4 - Symbols for Welding and Nondestructive Testing.
AWS D1.1 - Structural Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJune Road, P.O. Box 351040, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The sling spreader shall be as shown in figures 1 through 4, and as specified herein. When assembled, the sling spreader shall be adjustable from 7 to 10 feet in 3-inch increments for locking in position with toggle pins.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3 in accordance with 4.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The sling spreaders shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which sling spreaders may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.1.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the sling spreaders may be newly fabricated from recovered materials to the maximum extent practicable, provided the sling spreaders produced meets all other requirements

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of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the sling spreaders.

3.3.2 Tubing. Steel tubing shall conform to ASTM A 513, grade 4130, type and condition optional.

3.3.3 End plates. End plates shall be steel conforming to ASTM A 322, UNS G41300, condition optional.

3.3.4 Toggles. Toggles shall be brass conforming to QQ-B-637, copper alloy no. 464, half-hard temper.

3.3.4.1 Toggle pins. Toggle pins shall be steel conforming to ASTM A 36.

3.3.5 Chain. Steel chain shall conform to RR-C-271, type II, class 4, nominal thickness 0.065 inch.

3.3.6 Flathead pins. Steel flathead pins shall conform to MS20392, dash no. 12C109.

3.3.7 Washers. Steel washers shall conform to FF-W-92, type A, grade I, class A, natural finish.

3.3.8 Cotter pins. Cotter pins shall conform to FF-P-386, type B, class optional.

3.3.9 Finish. Surface and edge finish as shown in figure 2, 3, and 4 shall be in accordance with ANSI B46.1.

3.4 Performance. The assembled sling spreader shall lift a safe working load of 20,000 pounds and a proof load of 40,000 pounds without rupture or permanent deformation of any component.

3.5 Identification marking. The spreader shall be identified in accordance with, MIL-STD-130. Markings shall be 0.25 inch high and shall be die-stamped on the outer side of the center section.

3.6 Treatment and painting. The portions of the spreader normally painted shall be cleaned. treated, and painted in accordance with MIL-T-704, type F or G. The color of the finish paint shall be green 383 color #34094.

3.7 Workmanship. Components of the spreader shall not have burrs or projections which could prove injurious to using personnel. The spreader components shall not have cracks, splits, or breaks.

3.7.1 Welding. The surface of all parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welds shall be free from defects such as porosity, splatter, entrapped slag, overlap, undercut, lack of fusion, and cracks. Welding shall not be used as a repair measure. Interpretation of welding symbols in figures 1 and 3 shall be in accordance with AWS A2.4. Welding, procedures and welding qualifications shall be in accordance with AWS D1.1.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3 1 Examination. The first article sling spreader shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3 2 Test. The first article sling spreader shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1.

4.4.3 Test. Samples selected in accordance with 4.4.1 shall be tested as specified 4.5.2.

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4.5 Inspection schedule.

4.5.1 Examination. The sling spreaders shall be examined as specified herein for the following defects:

MAJOR

101. Dimensions or tolerances not as specified.
102. Materials not as specified.
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of materials, material finishes, or treatment.
106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the sling spreader.
107. Components missing or not as specified.
108. Treatment and painting not as specified.
109. Workmanship not as specified.

MINOR

201. Identification marking not as specified.

4.5 2 Proof load test. Extend the sling spreader to its maximum length and attach a 2 leg wire rope sling through the holes in the end plates. The two upper sling legs shall be of equal length and shall converge in a lifting ring or shackle. During application of the proof load, the angles between each of the upper sling legs and the spreader shall be 45 degrees, plus or minus 2 degrees, and shall angle between each of the lower sling legs and the spreader shall be 90 degrees plus or minus 2 degrees. The two lower sling legs shall be of equal length and shall be subjected to a gross weight of not less than 40,000 pounds. Apply a vertical hoisting force to the lifting ring or shackle and raise the spreader with the proof load until the load is free of any support. Raise and lower the loaded spreader to the floor five times and hold suspended for periods of 5 to 10 minutes for a total of 30 minutes. A hydraulically applied proof load may be substituted for the gross weight of equal force. After the proof load test examine the spreader for evidence of rupture or deformation. Nonconformance to 3.4 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105

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4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects.

110. Spreaders not reduced to minimum size as specified for level A, B, or C.
111. Boxes not as specified for level A or B.
112. Quantities packed together exceed the weight limitation of the box for level A, B, or C.
113. Marking missing, illegible, incorrect, or incomplete for level A, B, or C.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The spreaders shall be reduced to minimum size without disassembly. The spreaders shall then be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, class 2, style optional, in quantities not to exceed the weight limitation of the box. The box shall be closed and strapped in accordance with the appendix to the applicable box specification, except the strapping shall be finish B, size as applicable.

5.1.2 Level B. The spreaders shall be packed as specified in 5.1.1 for level A except boxes shall be domestic type or class, as applicable and the strapping may be finish A.

5.1.3 Level C. The spreaders shall be reduced to minimum size without disassembly. Packing shall be in accordance with MIL-STD-1190.

5.2 Marking. Marking shall be in accordance with MIL-STD-129, (see 6.4).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The spreader is intended for use with wire rope slings employed to lift vehicles weighing up to 10 tons and is intended to eliminate contact between the sling and the sides of the vehicles.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval and the number of units required (see 3.2).
d. Level of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in

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acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Levels of preservation. Levels of preservation have not been included as the item needs no preservation. For purposes of packing level marking the preservation shall be considered level A.

6.5 Subject term (key word) listing.

Sling
Spreader
Telescopic sling spreader

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 AQL certification. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQL's/LTPD's (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

(Certified by/date:_____).

Custodian:
Army - ME

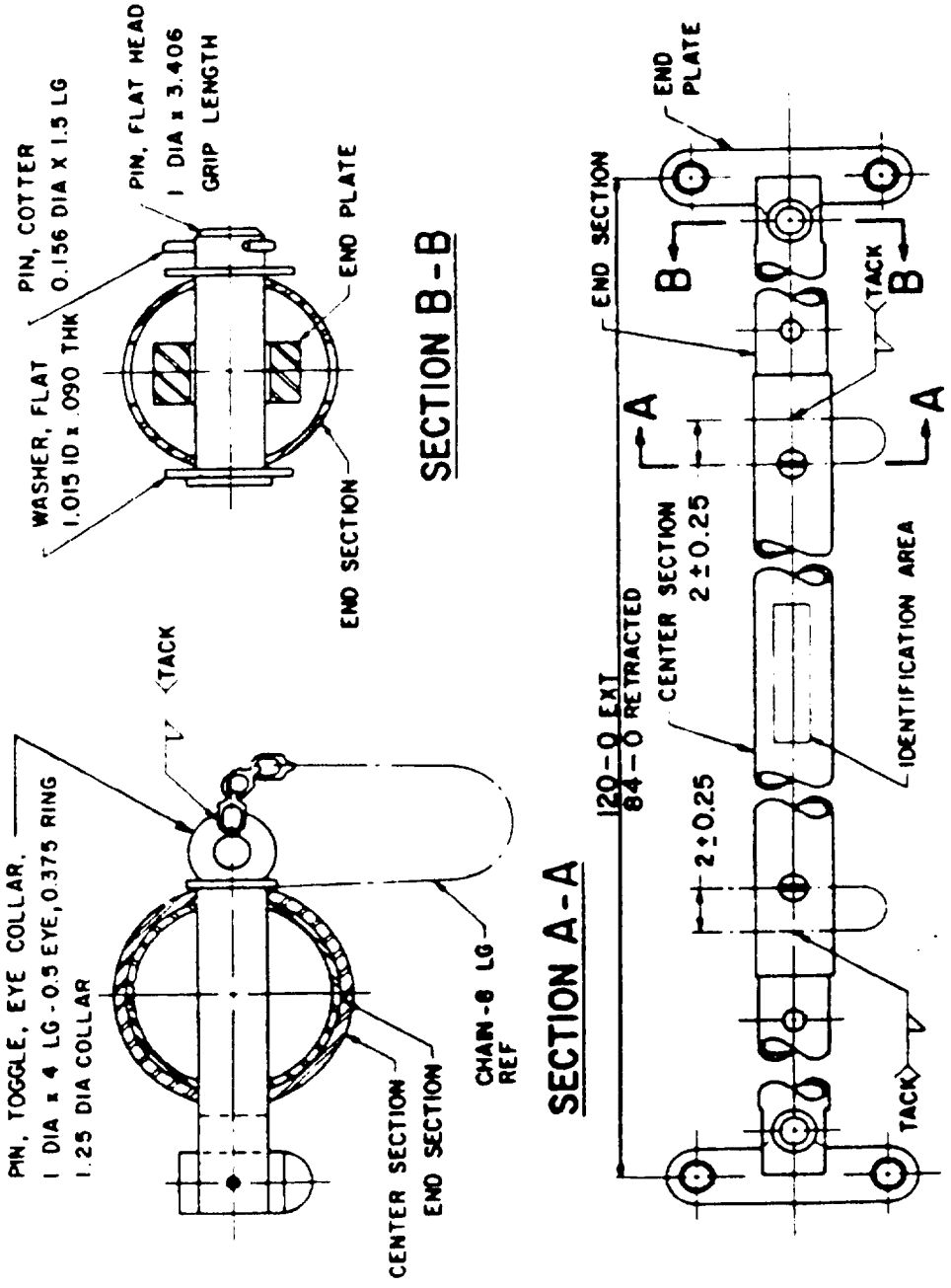
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Review activities:
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DLA - GS

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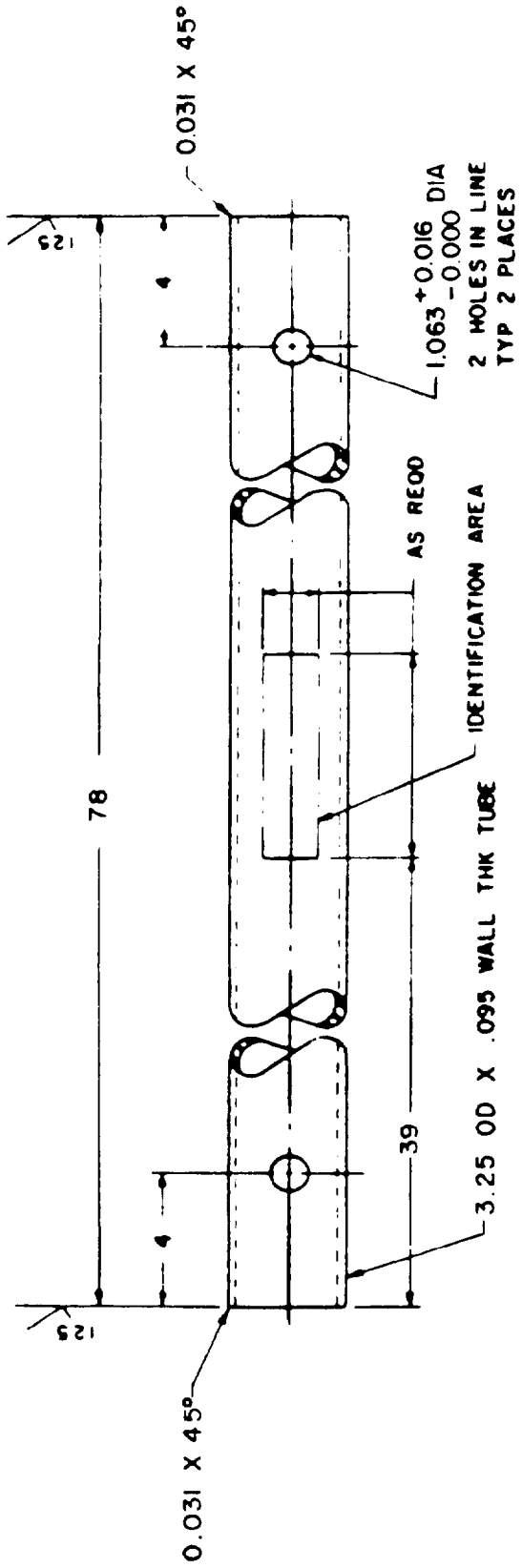


NOTE: ALL DIMENSIONS ARE IN INCHES.

FIGURE 1. Telescopic spreader bar assembly.

X-2901A

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X-2902A

FIGURE 2. Center section.

NOTES:
1. TOLERANCE UNLESS OTHERWISE SPECIFIED ± 0.062.
2. ALL DIMENSIONS ARE IN INCHES.

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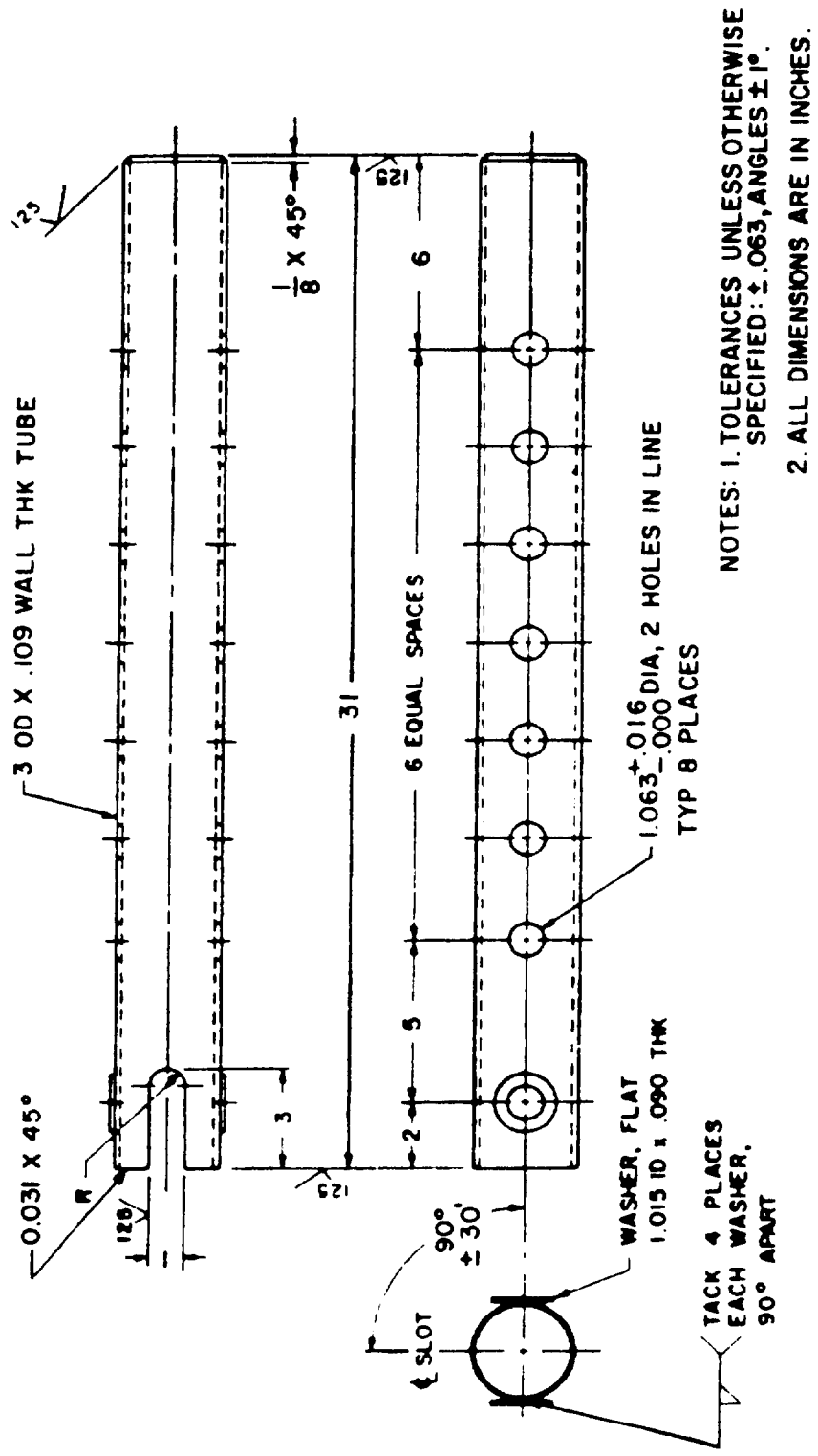
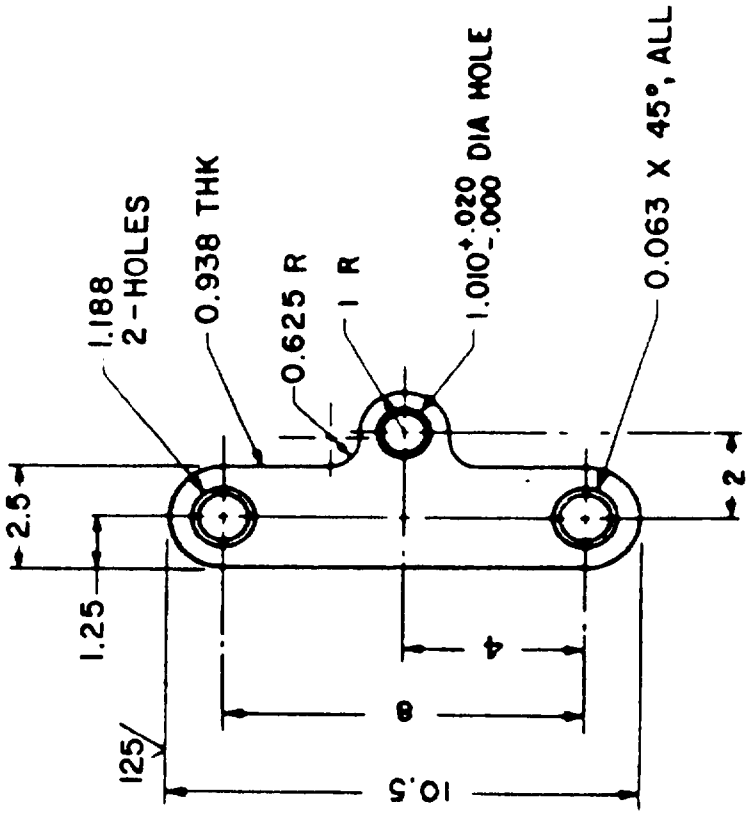


FIGURE 3. End section.

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NOTES:
1. TOLERANCES UNLESS OTHERWISE SPECIFIED ± 0.063, ANGLES ± 1°.
2. ALL DIMENSIONS ARE IN INCHES.

FIGURE 4. End plate.

X-2904A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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	MIL-S-52408C(ME)	891219

3. DOCUMENT TITLE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (703) 664-5717	(2) AUTOVON 354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft Belvoir VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340	AUTOVON 289-2340

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